

EEF Meeting Minutes, Thursday Sep 29, 2022

Erlang Ecosystem Foundation, Minutes of a Regular Meeting of the Board of Directors Meeting #71, September 29, 2022

A regular meeting of the Erlang Ecosystem Foundation was held over group conference call (Zoom). The meeting began at 16:03 UTC.

Attendance

The following members of the board of directors were present (11/11): Sophie DeBenedetto, Francesco Cesarini, Maxim Fedorov (had to leave at :45, after item 5.3), Kenneth Lundin, Ben Marx, Bryan Paxton (joined at :15, after item 4), Miriam Pena Villanueva (joined after item 5.2), Frédéric Trottier-Hébert, Peer Stritzinger, Sebastian Strollo, and Alistair Woodman.

The following members of the board of directors were absent: [none]

Agenda Items

1. Approval of minutes from previous meetings.

RESOLVED, that the Erlang Ecosystem Foundation approve the minutes as presented as representing a true and accurate record of the board meeting #70 held on Sep 15, 2022. Unanimously approved.

2. Intermediate actions: none to report

3. No financial update. Membership update: Annual supporting 134, Basic membership 1050, Lifetime Supporting 18, Managing and Contributing 15.

4. Unanimous consent votes: none scheduled

5. Stipend applications:

1. Automated Checking of Refactoring Correctness (to be integrated with Wrangler, and thus usable via the Erlang Language server using the Wrangler plugin), development support at €5000 (to pay a full or part-time student developer), submitted by Simon Thompson.

RESOLVED, that the Erlang Ecosystem Foundation approve the stipend application for development of an Automated Checker of Refactoring Correctness at

€5000. Vote: 9 for, none against. Maxim abstained. Miriam had not yet joined the meeting.

2. Travel Stipend to attend Code BEAM America, submitted by Dominik Katkó for \$750 for helping with travel costs to be able to present “Refactoring via the Language Server Protocol” (a project sponsored by the EEF).

RESOLVED, that the Erlang Ecosystem Foundation approve the stipend application from Dominik for \$750 to support travel to present “Refactoring via the Language Server Protocol” at the Code BEAM America conference. Vote: 9 for, none against. Francesco abstained. Miriam had not yet joined the meeting.

3. Static Type Checker, from Annette Bieniusa at TU Kaiserslautern. For \$25000 for 4-5 man months of development work over a 6 month period. Some discussion and open questions about the project - Peer to forward questions to Annette. No decision at this point.
6. Annual General Meeting 2022 planning. To be held in conjunction with CodeBEAM America (first week of November). Discussion around what we would like from working groups (to share at the AGM).
7. Any other business: none
8. Alistair declared the meeting adjourned at 16:54 UTC.

Dated September 29, 2022

Secretary of the meeting: _____ Sebastian Strollo

Attachments:

Travel stipend application (1 page)

Stipend application for “Automated Checking of Refactoring Correctness” project (2 pages)

Stipend Type : conference

First name : Dominik

Last Name : Katkó

Email Address : katkodomini@gmail.com

City : Copenhagen

Country : Denmark

Bio : Computer Science and Engineering MSc. student at Technical University of Denmark

Purpose: I am a co-speaker (with Simon Thompson) in a talk called "Refactoring via the Language Server Protocol" in Code Beam America on 2022 Nov 3-4. Details: <https://codebeamamerica.com/talks/refactoring-via-the-language-server-protocol/> The talk is based on my work to update and incorporate refactoring facilities from Wrangler into the Erlang Language Server. This work was also supported by EEF, and we agreed with Simon that I will further improve the Wrangler Language Server until and after the conference. I undertook this work as an undergraduate at ELTE Budapest, and I am now studying for a master's in Denmark. This trip will be my first international conference visit, and I look forward not only to giving the presentation itself but also to networking with the Erlang and Elixir communities at the meeting. That is why I am applying for funding to cover my stay during the conference (Nov 2-5). My flight expenses will likely be covered by Erlang Solutions.

Report : The talk will publicise my work, and the Erlang Language Server in general, and should drive adoption of the tools, as well as contributors to further improve these tools and Wrangler itself. We are at least expecting bug reports which we can work on afterwards. The feedback and contribution from the Erlang community will be written in the report as well as our further work improving Wrangler.

Stipend Amount (USD) : 750

Beneficiaries : I will be the beneficiary of the stipend. I will spend the money on hotel costs and subsistence (approx 250\$/day)

EEF project proposal

Simon Thompson, September 2022

Automated Checking of Refactoring Correctness

Refactorings are program transformations that are intended to improve code in some way, e.g. by making it scalable, more efficient or more readable, without changing its behaviour. Tools that automate refactorings can therefore present a critical weakness in the software engineering workflow, in that they can *reduce* the security of code, and so it is crucial to give the strongest possible assurance to users of these tools. The state of the art currently is to deploy regression testing to check that none of the acceptance tests for code are broken by the refactoring, but that gives only a weak guarantee.

These projects aim to advance beyond the state of the art to develop a suite of techniques that can be used to verify the correctness of refactorings by automatically checking the equivalence of programs before and after refactoring, and in the case that equivalence cannot be established, to provide concrete evidence of how the transformation fails to be correct. The tool would be equally applicable to refactorings performed by automated tools and to those done “by hand”.

There is some literature based on program equivalence checking in general, the work proposed here is innovative in focussing particularly on systems before and after refactoring, rather than arbitrary pairs of systems. The case of refactorings is special in that the two systems share much of their structure, and this structure can be used to guide the verification, or indeed the search for a counterexample by focussing on their structural differences. For this reason we expect to be able to make considerably more progress in automation than in the case of general equivalence checking.

One way of checking function equivalence is to use testing, and a recent example of that is given in this paper: https://link.springer.com/chapter/10.1007/978-3-030-99429-7_11. This builds on earlier work on context-aware, localised equivalence verification, as described in the PEQCheck paper <https://arxiv.org/abs/2101.09042>.

Project: Implement refactoring correctness checking for Erlang

The aim of this project is to build, in Erlang, an open source system for checking the equivalence of Erlang code before and after a refactoring is performed, and in the case of failure to exhibit concrete counterexamples. The result should be incorporated into the Wrangler refactoring tool, and thus with Erlang Language Server (via the Wrangler Language Server).

Details

In stages, the project team should

- Understand the work in the two papers referenced above, and to compare it with other work in the area, e.g. using QuickCheck and Proper, property-based testing tools for Erlang. [1 month full time]
- On this basis, design the overall system and data structures and algorithms within it, and then implement the result. [2 months]
- Evaluate the effectiveness of the system on a range of automated and “by hand” refactorings. [1 month]
- Finalise documentation and write-up. [1 month]

The deliverables of the project will be

- A write-up of the results in a form that can be presented to a developer conference and published in the peer-reviewed literature.
- The code itself will be available through a suitable open source github repository.
- Documentation, including architecture description and API documentation.

Costing: €5000k @ €1000k/month; pay for full or part-time student developer.

Supervisory team: Simon Thompson (Eötvös Loránd University and University of Kent),
Dániel Horpácsi (Eötvös Loránd University).